# Keshav Sharma

India | 8168289407 | keshavsharma71982@gmail.com | LinkedIn Link | GitHub Link | Portfolio

## EDUCATION

**Vellore Institute of Technology** — **Chennai, Tamil Nadu**

*Bachelor of Technology in Computer Science (Data Science Specialisation)* — *Graduation: May 2027*

## SKILL SUMMARY

**Languages:** Java, C/C++, Python, SQL, JavaScript, R
**Software Engineering & AI:** Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Database Management, Git, Machine Learning (NLP, CNN, Transformers)
**Frameworks & Libraries**: React.js, Node.js, Scikit-Learn, Pandas, NumPy, Matplotlib, TensorFlow, Keras
**Development Tools:** Git, VS Code, IntelliJ, Jupyter Notebook, SQLite, Power BI, Tableau

## PROJECT EXPERIENCE

***AI-Powered Expense Receipt Scanner & Analytics Platform*** **|LINK**

- Designed and implemented a comprehensive full-stack expense management platform that automates receipt processing; enhanced processing time by 40% while ensuring accuracy through EasyOCR, OpenCV, and advanced NLP-based parsing techniques.
- Implemented and deployed a scalable application using FastAPI, React, Vite, SQLAlchemy, SQLite, JWT Authentication, and Google OAuth, featuring expense analytics, budget tracking, category-wise spending insights, receipt management (CRUD), Excel export, and a responsive dashboard hosted on Render and Vercel.

***Real-Time Stock Anomaly Detector*** **| LINK**

- Built an end-to-end anomaly detection pipeline on 2 years of NSE stock data, engineering 9 financial features (RSI, Bollinger Bands, volume ratio); deployed dual models — Isolation Forest (2.01% anomaly rate) and LSTM Autoencoder — achieving temporal pattern detection across 448 trading sessions with an interactive Streamlit dashboard
- Integrated FinBERT financial sentiment analysis on live news headlines to correlate price anomalies with news tone, identifying high-confidence anomaly events such as the Apr 2025 market selloff (score −0.6332) and Jan 2026 peak reversal through cross-validation against NSE announcements

***Customer Lifetime Value Engine*** **|LINK**

- Developed an end-to-end Customer Lifetime Value (CLV) engine using BG/NBD and Gamma-Gamma probabilistic models on 1M+ retail transactions, forecasting 12-month CLV for 4,000+ customers with churn-risk scoring and value-based segmentation.
- Developed an interactive Streamlit retention analytics platform with a campaign ROI simulator and Power BI dashboard, enabling data-driven targeting of high-value at-risk customers through probabilistic purchase and survival modelling.

***Docwise*** **|LINK**

- Engineered Docwise's AI-powered PDF Q&A system using Google Gemini, enabling high-accuracy (95%+) document querying and reducing manual search time by 40%.
- Implemented TF-IDF vectorisation, chunking, and real-time processing pipelines for PDFs exceeding 200+ pages, improving system scalability and responsiveness.
- Designed and integrated chat history and session management in Streamlit, boosting usability and user retention by 35%.

## Certifications

**Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | |LINK** ***2026***

- Demonstrated proficiency in Generative AI, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, responsible AI practices, and building AI applications using Oracle Cloud Infrastructure (OCI).

**Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate |LINK** ***2026***

- Demonstrated foundational knowledge of AI, Machine Learning, Generative AI, Large Language Models, and Oracle Cloud AI services.

**VOIS – Vodafone Idea Foundation | LINK** ***2025***

- Virtual Internship (AICTE-certified) — LLM-based data analysis on Airbnb/Netflix datasets

**BCG Data Science Job Simulation |LINK** ***2025***

- Performed customer churn analysis using Python (Pandas, NumPy), built and optimised a Random Forest model achieving 50% recall, and delivered data-driven business recommendations through visualisations and executive reporting.